Database Lab : Relational Schema Architecture, Constraints & Indexing

Duration : 30 Min.

Scenario

As a Database Engineer at NIIT, you are tasked with creating a normalized relational schema in PostgreSQL enforcing primary key, unique, and foreign key referential integrity constraints, and creating a secondary B-Tree index on foreign key columns.

Input Details

The environment has been pre-configured with the following local database platform instances:

- **Database Host:** `localhost` (Ubuntu Linux VM)
- **PostgreSQL Service:** `localhost:5432` | DB: `postgres` | User: `postgres` | Password: `PostgresPassword123!` | CLI: `psql`

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (e.g. `student-exam123`).

Resource naming: - **Customers Table:** `<prefix>_customers` - **Orders Table:** `<prefix>_orders` - **Foreign Key Index:** `idx_<prefix>_orders_cust`

Task Objectives

Perform the following relational DDL steps directly in PostgreSQL using `psql`:

1. Create Customers Table

- Create table `<prefix>_customers` with columns:
- `id SERIAL PRIMARY KEY`
- `email VARCHAR(100) UNIQUE NOT NULL`
- `full_name VARCHAR(100) NOT NULL`
- `created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP`

2. Create Orders Table with Foreign Key

- Create table `<prefix>_orders` with columns:
- `order_id SERIAL PRIMARY KEY`
- `customer_id INT NOT NULL REFERENCES <prefix>_customers(id)`
- `amount NUMERIC(10,2) NOT NULL`
- `order_status VARCHAR(20) DEFAULT 'PENDING'`

3. Create Secondary B-Tree Index

- Create a secondary B-Tree index named `idx_<prefix>_orders_cust` on `<prefix>_orders(customer_id)`.

4. Verify Constraints with Seed Data

- Insert a customer record, insert an order referencing that customer, and verify that foreign-key integrity prevents inserting an order with a non-existent `customer_id`.

Verification

Once you have created the tables and index, run the verification script:

```
python run.py
```

The verification system inspects `information_schema` and `pg_indexes` on `localhost:5432`.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Primary Key & Unique Constraint Definitions (_customers)	4 Marks
TC2	Foreign Key Referential Integrity (_orders -> customers)	3 Marks
TC3	B-Tree Index Creation (idx__orders_cust on customer_id)	3 Marks

Total Score: 10 Marks

Important Notes

- Constraints must be enforceable in PostgreSQL catalog.
- Replace `<prefix>` with your lowercase candidate prefix.